

EXERCISE SHEET 14

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

14.1. Find rational canonical forms of the following matrices over $\mathbb{Q}$:

$$\begin{aligned} \text{(a)} \quad A &= \begin{pmatrix} 1 & 5 & 7 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}; \\ \text{(b)} \quad B &= \begin{pmatrix} 2 & 4 & 0 \\ 1 & 4 & 1 \\ 3 & 8 & 3 \end{pmatrix}. \end{aligned}$$

EXTRA QUESTIONS

14.2. For a ring R and M an R -module, let $T(M) = \{x \in M \mid \exists r \in R \setminus \{0\} \text{ such that } rx = 0\}$. Find $T(M)$ for the case $R = M = \mathbb{Z}/6\mathbb{Z}$. Is the set $T(M)$ here a submodule of $\mathbb{Z}/6\mathbb{Z}$?

14.3. Let V be a 3-dimensional vector space over $\mathbb{Q}$ and let $T : V \rightarrow V$ be a linear transformation. Suppose the matrix of T with respect to the standard basis is the matrix A given in Exercise 14.1(a). Show that V is a T -cyclic space and find the minimal polynomial of T on V .

14.4. Find the rational canonical form for $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $a \neq 0$, viewed as a matrix over $\mathbb{R}$.